Nissai Glacier — Field Card, September 2026

Sangvor valley, Tajikistan · Cromo-Adapt / PAMIR · WSL-SLF + CRG-TAS · AWS at 71.35392 E, 38.74287 N, 3909 m · essentials only — full protocol held separately

MUST BE IN THE BAG — every item in red is the **only** route to an objective. If it is not in the bag, that objective is lost for the year.

> **ALPINE STYLE — DO NOT OVER-PACK**
>
> Work at the site is **~1.5 days**. Pack for that plus one contingency day, **not a week**. Food and camp weight are the easiest things to over-carry and the hardest to justify on the walk in — everything here is carried up ~1480 m from the roadhead. Weight spent on food is weight not spent on the auger, the stake segments and the spares, which are what the trip is actually for.

PACK LIST — TICK AT LOAD-OUT, TICK AGAIN AT DUSHANBE

READOUT & COMPUTING

- [ ] Toughbook, charged · charger · 12 V/inverter lead
- [ ] LoggerNet installed and opened once
- [ ] **CR300-series USB driver pre-installed** — no internet beyond Sangvor
- [ ] **2× USB cable for CR350** — A-to-C if the laptop is USB-A
- [ ] **433 MHz GeoPrecision dongle** — no substitute exists
- [ ] **GP5W-Shell installed and tested with the dongle**
- [ ] Laptop clock settable to UTC, DST off

DATA MEDIA — THE TWO-COPY RULE

- [ ] **2× external media**, carried by two different people
- [ ] Spare SD card — the AWS sonic year lives only on the card
- [ ] 256 GB timelapse card (~70 GB/yr)
- [ ] Card reader + spare cables

AWS SPARES & SERVICE

- [ ] **AWS logger box key** — held by Mark; the enclosure cannot be opened without it
- [ ] Battery · desiccant · fuses · cable · mounting hardware
- [ ] Cleaning kit — radiometer domes, SI-111 window, SHM31 optical window
- [ ] Timelapse camera batteries (replace regardless of level)

SAFETY & COMMS

- [ ] **Satellite phone** + spare battery — no phone coverage beyond Sangvor
- [ ] Walkie-talkies, one per party, charged + spares
- [ ] **First aid kit**
- [ ] Agreed radio channel and check-in times, written down

STAKE DRILLING

- [ ] **Kovacs 6 m auger + drill**
- [ ] 4 drill batteries
- [ ] **Stake segments** + chain / screws — **carry the shortfall only.** ~6 were left on site in 2025; confirm the count and condition with Jovid before load-out
- [ ] 2× tape measure

SURVEY & RECORDING

- [ ] **2× GPS**, both preloaded with the geo package and the stakes GPX
- [ ] Camera + spare batteries
- [ ] Field sheets printed · pencils (not pens)
- [ ] This card, printed

GROUND-TEMPERATURE NETWORK (only if the BLE network is being read)

- [ ] **Readout phone on Android 13 or older** with SLF-Logger Manager installed — Android 14+ refuses the APK
- [ ] Do **not** factory-reset a phone that already has it

GLACIER TRAVEL — 1 ROPE TEAM OF 4

- [ ] **Glacier rope**, 1× 30–50 m
- [ ] 4× harness · 4× helmet
- [ ] 4× crampons · 4× ice axe
- [ ] Crevasse rescue per person: 2× prusik or ascender, pulley, 2× locking carabiner, sling
- [ ] 4–6 ice screws
- [ ] Snow stakes / deadman

Before departure — chase, decide, confirm

Nothing below can be fixed once the road ends at Nissai. Sangvor has 4G; beyond it there is none.

BEFORE ANYTHING ELSE

Satellite telemetry has been dead since 15 March 2026 (Astrocast outage; vendor unreachable). Nobody has seen the AWS in ~6 months — **arrive expecting to repair, not service**.
Rockfall: two large falls from the north wall south of basecamp, 2 Sep 2025 ~15:00. Reassess runout before siting camp.
Ridge block slope is off-limits — moving as the glacier toe melts out. WP100 is read by radio from the glacier instead.
Confirm the Nissai stream bridge by phone before departure (Sangvor has 4G, Nissai none); commission a rebuild in advance if needed.

BLOCKING — THESE STOP DATA BEING COLLECTED OR INTERPRETED

	ITEM	OWNER	WHY IT BLOCKS
☐	**Sensor depth register** glacier ice station	Evan Miles (Stefan / Ines)	Without it the recovered temperatures **cannot be assigned to depths** — the year's data is uninterpretable. Also need coordinates, the 2025 summary, and emergent mast height at install.
☐	**CR350 security code / PakBus key**	Evan Miles	If a code was set in 2025 the logger **will not respond and will read as dead hardware**. Defaults are address 1, code 0, key blank.
☐	**Astrocast / AstroNode decision**	Andi	Telemetry dead since 15 Mar 2026, vendor unreachable and possibly out of business. If a replacement path is fitted in 2026 **the hardware travels with the team**.
☐	**Glacier station replacement decision**	Joel / Evan	If the station is found destroyed (considered likely) and brought down — is it replaced this season? Replacement hardware travels in **on the same trip or not at all**.
☐	**Stake cache on site** how many, what length, what condition	Jovid	~6 stakes were left at the site in 2025. Assume all 7 need redrilling, so the question is only **how many segments to carry up**. Over-carry is dead weight on a 1480 m ascent; under-carry loses the stake network for the year. **Get a number before load-out.**
☐	**AWS logger box key** collect before departure	Mark	The AWS enclosure **cannot be opened without it** — no download, no SD card, no service. Activity 1 does not happen. Confirm who is physically carrying it.
☐	**GP5W-Shell + 433 MHz dongle** tested together, on the actual laptop	Field team	Without the dongle **WP100 cannot be read at all**, and it is now the only sensor reporting from the suspended ridge.
☐	**Readout phone** only if the BLE network is read	Field team	SLF-Logger Manager targets API 19; **Android 14+ refuses to install it**. Take a phone on Android 13 or older, or prove `adb install --bypass-low-target-sdk-block` works on the actual handset.

CONFIRM BEFORE LEAVING — NOT BLOCKING, BUT CHEAP NOW AND EXPENSIVE LATER

	ITEM	DETAIL
☐	**Nissai stream bridge**	Phone from Sangvor (4G) and confirm the foot bridge is standing. **Commission the rebuild in advance if not** — the bridge team is engaged from Sangvor.
☐	**WP100 model + access code**	Confirm M-Log5W-Rock and that access code `DE10` is still current. Logger ID `A542D3`.
☐	**Geo package on both GPS units**	Full package (23 points) plus the stakes-only GPX. Export a fresh GPX every evening in the field.
☐	**WP54 — never read since 2021**	4616 m, hard access. **Decide before departure** whether it is attempted this season.

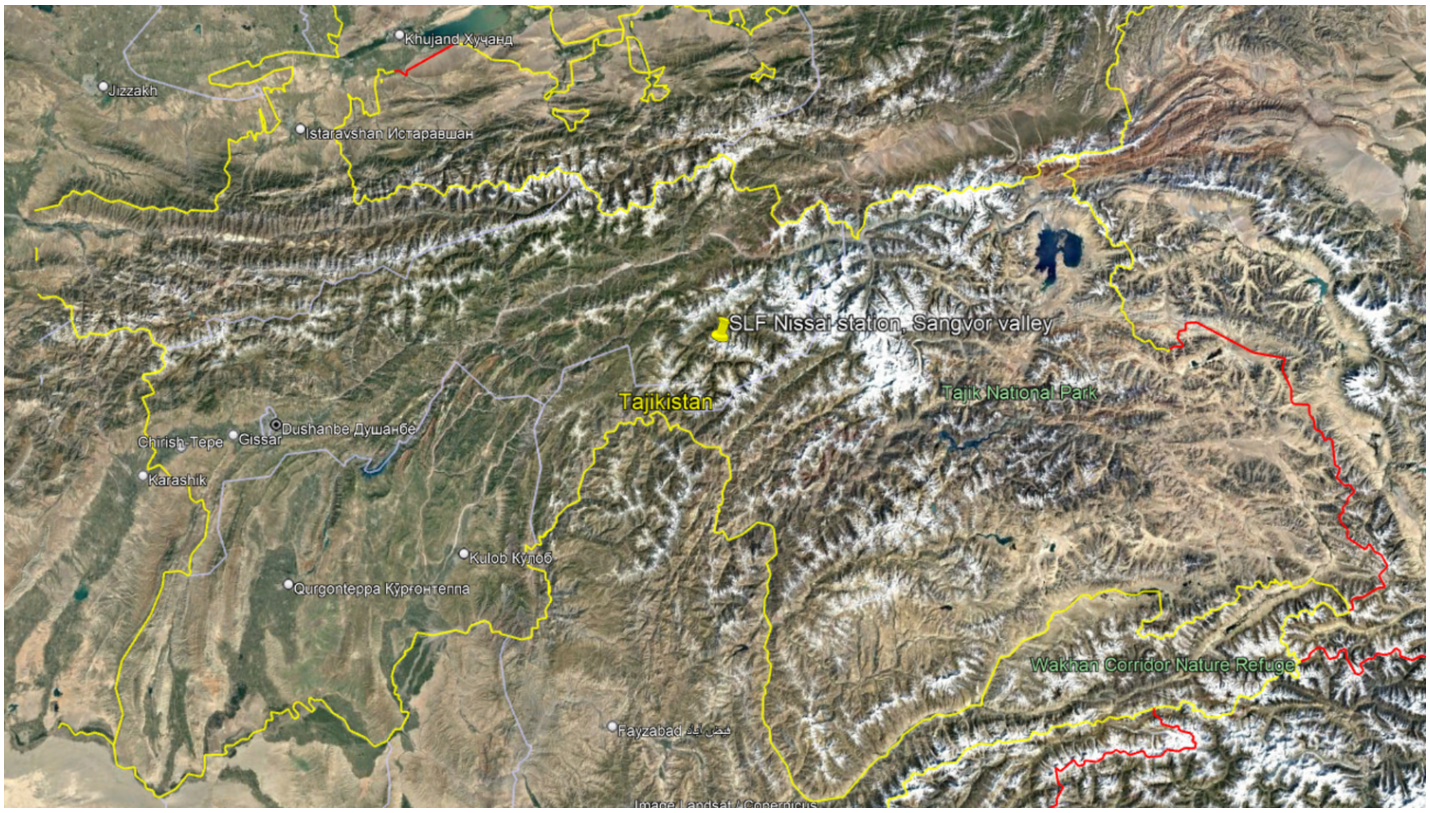
RESOLVE IN THE FIELD

☐	**NI 6-25 elevation**	4343 m (2025) vs 4383 m (2024) at a near-identical position — **re-survey and flag**.
☐	**NS8-24 · the 107 cm reading**	Appears against both the 2024 and 2025 columns in the source tables. Establish which year it belongs to.
☐	**Stream gauge**	The 2025 report appears to carry the 2024 text forward unchanged, so what is on the ground is unknown. Not covered by this card — see the long protocol.

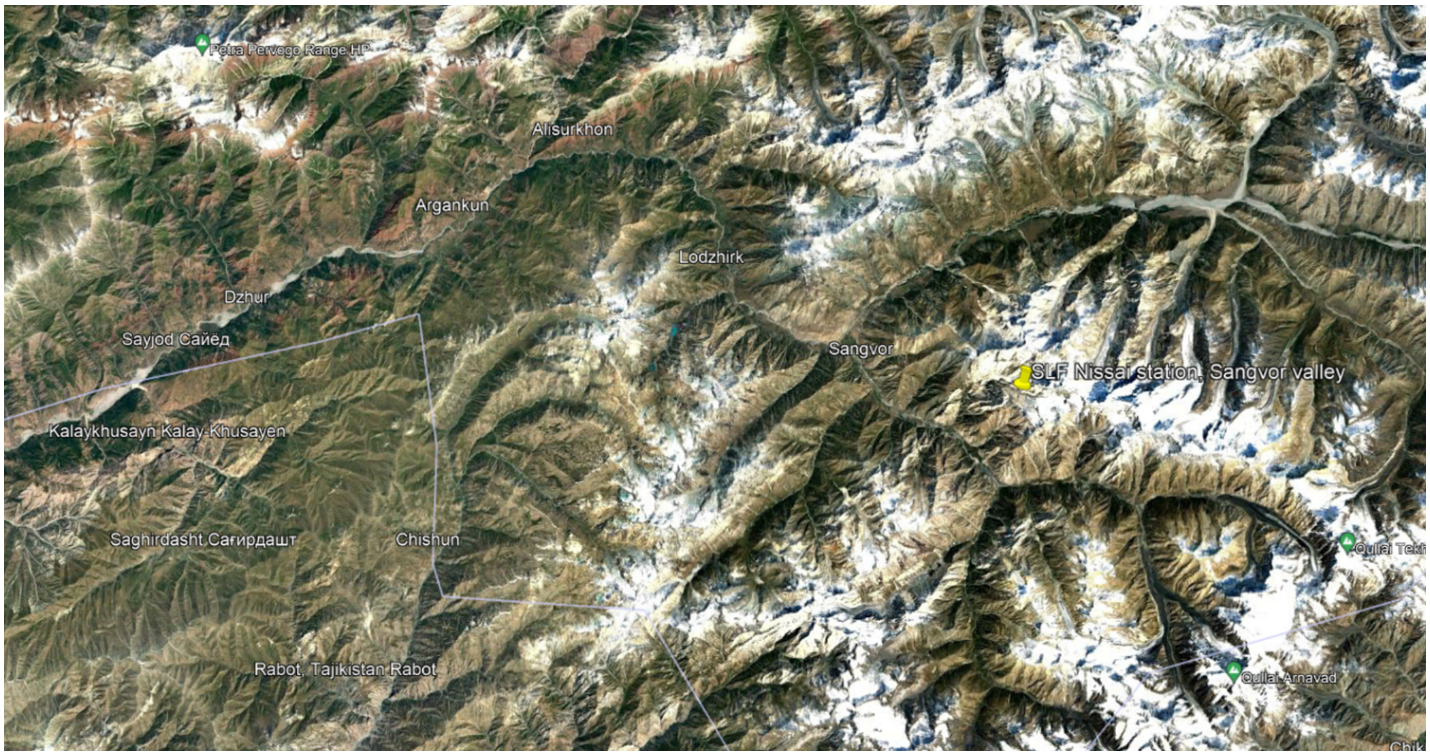
CONVENTIONS — APPLY EVERYWHERE

ITEM	RULE
Lengths	**Centimetres, whole numbers.** Never mix metres onto a field sheet.
Coordinates	Decimal degrees WGS84, 6 dp, longitude (X) then latitude (Y).
Clocks	**Record the timezone actually set, every time.** CR350/AWS: local UTC+5. GeoPrecision: **UTC**. An unrecorded offset is invisible in the data.
Waypoints	New GPS waypoint at every stake, logger and station — even when re-occupying. Export GPX daily.
Photos	As found · as left · any damage. Note frame numbers.
Every evening	**Copy the day's data to two media carried by two people.** Transcribe field sheets the same night.

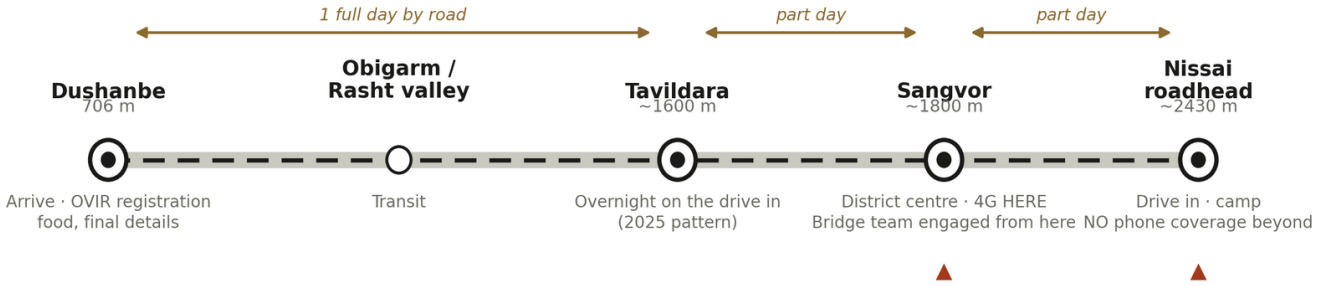
M Maps — journey



1 · The site within Tajikistan. SLF Nissai station, Sangvor valley, West Pamir. *From the 2025 field report, Figure 1.*



2 · Approach. The site relative to Argankun, Sangvor and the Petra Pervogo Range. *From the 2025 field report, Figure 2.*



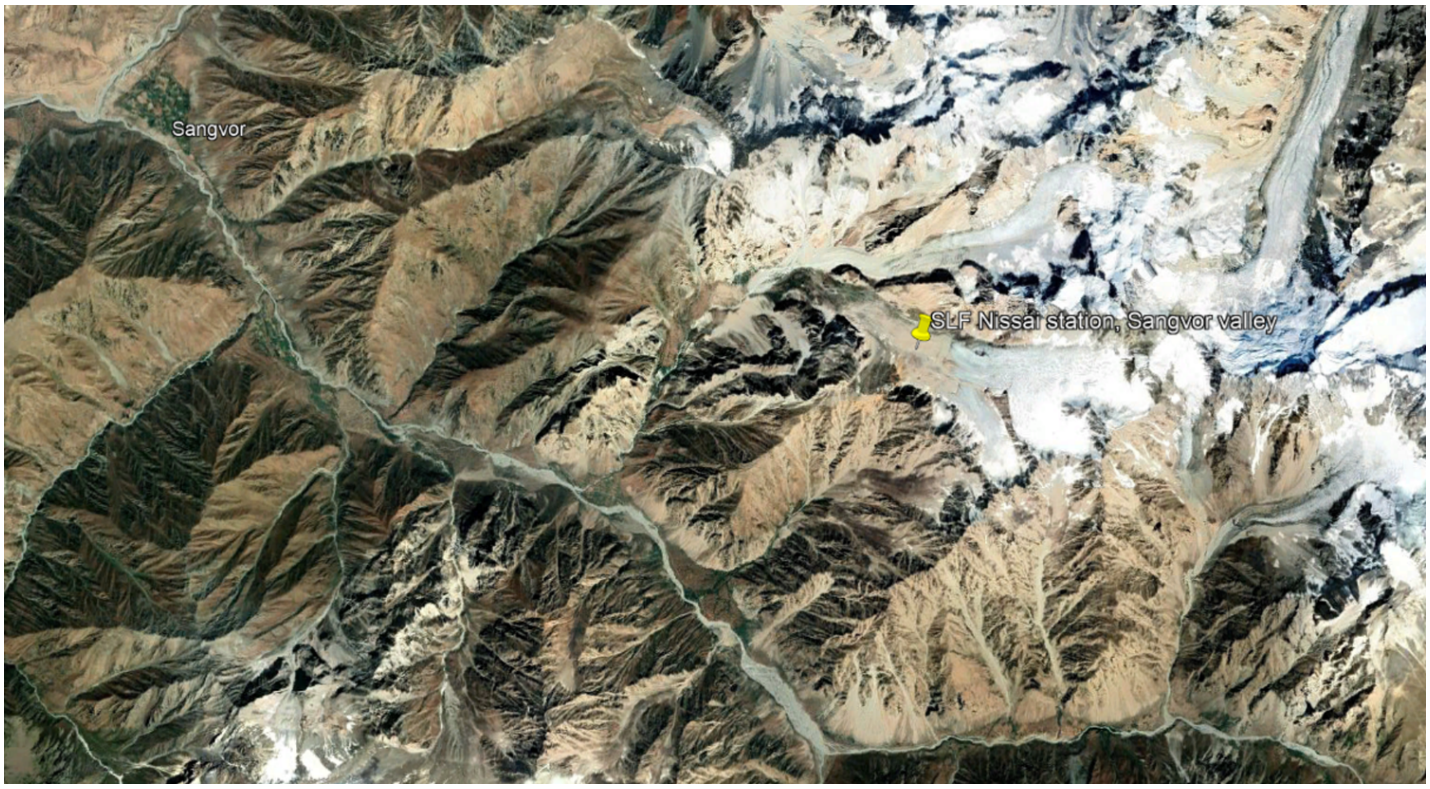
▲ At Sangvor — CALL AHEAD: confirm the Nissai foot bridge is standing, and commission its rebuild in advance if not.

▲ At Nissai — the road bridge (Oct 2022) avoids fording the Nissai stream on the drive in.

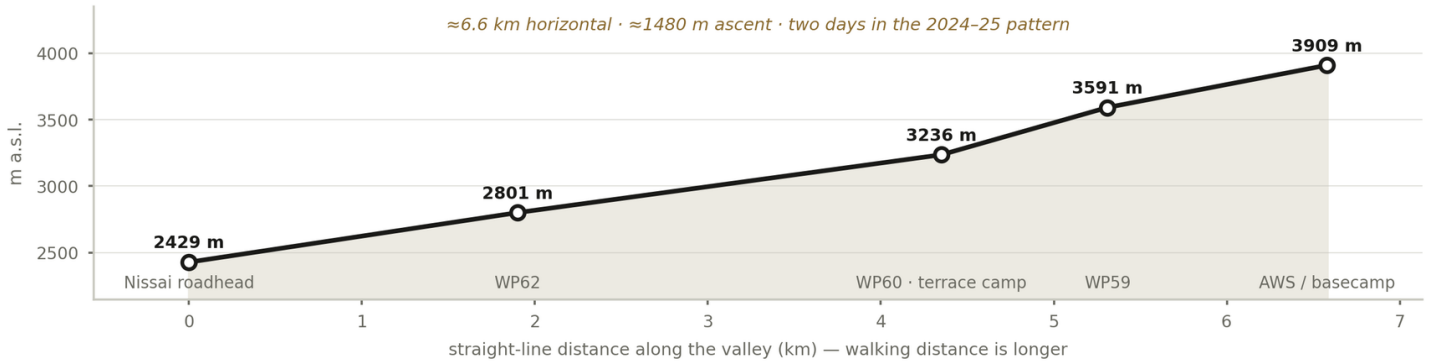
Schematic — sequence and overnight stops, not to scale. Road distances not verified; durations from the 2024–25 field reports.

3 · Journey sequence. Overnight stops and the two things to act on en route. Schematic, not to scale.

M Maps — hike in



4 · Nissai valley, Sangvor to the station. *From the 2025 field report, Figure 3.*



5 · Elevation profile along the valley, from the surveyed ground-temperature logger positions. Terrace camp and basecamp are approximate; logger positions are surveyed.

1 Automated weather station

WP51 · 71.35392 E, 38.74287 N · 3909 m · Campbell CR1000X · established 2021. Half a day.

> **SD CARD — THE YEAR'S SONIC RECORD LIVES ONLY HERE**
> The sonic anemometer's full high-frequency record is on the **SD card only**, not in logger memory. Press **eject** (flushes memory to disk), then **wait for the LED to turn GREEN before pulling the card** — about 30 s. Pulling early truncates or corrupts the year.

PROCEDURE

1. **Photograph before touching anything** — mast lean, ice accretion, guy tension, panel soiling, animal damage.
2. **Diagnose the AstroNode first.** Photograph the antenna and its sky view. Check for rime, obstruction, orientation and level. Read the AstroNode's own status / queue and record whether it believes it is transmitting. *This is the only evidence separating a station-side fault from the vendor outage.* Clear and re-aim; photograph as left.
3. **Diagnose the SHM31** (snow depth dropouts). Clean the optical window; confirm nadir aim and rigid mounting; inspect the target surface for poor reflectivity (dark debris, meltwater, ice glaze). **Take a manual tape measurement from sensor face to surface and record it with a simultaneous sensor reading** — that offset is what lets the record be corrected later.
4. **Record station state:** battery voltage, logger clock drift vs GPS (*record before correcting*), desiccant colour, water ingress.
5. **Download logger memory.** Verify the file opens and the last record is today.
6. **Eject and copy the SD card** per the box above. Copy to **two** separate media before it leaves your hand. Verify record count and continuity *on site*.
7. **Reinsert, confirm logging resumes**, watch one scan complete.
8. **Service sensors:** clean radiometer domes and SI-111 window; clear SHM31 path; check FlowCapt tubes clear and mounting rigid; confirm sonic transducer paths unobstructed.
9. **Timelapse camera:** swap the 256 GB card (~70 GB/yr). **PIR stays OFF.** Keep the 2024 acquisition schedule; record it as set. Replace batteries regardless of level.
10. **Photograph as left. Write down every change made** — an undocumented change becomes an unexplained step in the timeseries.

Date / time on site		Observers	
Battery voltage (V)		Logger clock drift (s)	
Last record timestamp		SD green LED seen?	
SD copied to 2 media?		Desiccant / ingress	
Antenna iced / obstructed?		Antenna orientation	
AstroNode status as read			
SHM31 window / target surface		Tape (cm) / sensor reading (cm)	
Timelapse card + schedule set		PIR confirmed OFF	
Damage, anomalies, changes made			

2 Glacier ablation stakes

Nissai Glacier · 7 ablation stakes 4068–4387 m + accumulation stake NS8 · budget a full day on the ice.

Convention. *Stake measured* = **emergent length above the ice surface**, cm. *Stake in ice* = length below surface at drilling. So emergent = length − in ice.

ablation (cm i.e.) = emergent$_{2026}$ − emergent$_{2025}$ · b (m w.e.) = − cm i.e. × 0.9 / 100 (ρ_{ice} = 900 kg m^{-3})

Check, stake 6 in 2024–25: length 400, in ice 228 → emergent 172. Measured 378. Ablation = 206 cm i.e. → −1.854 m w.e.

AS LEFT SEPTEMBER 2025 — WHAT YOU SHOULD FIND

STAKE	LABEL	LON (X)	LAT (Y)	ELEV	LENGTH	IN ICE	EMERGENT	24–25 ABL	PROJECTED SEP 2026
1	NI 1-25	71.362939	38.741813	4068	600	595	5	481	~486 emergent — **near melt-out**
2	NI 2-25	71.365724	38.739660	4127	600	600	0	492	~492 emergent — **near melt-out**
3	NI 3-25	71.369215	38.739268	4234	600	600	0	370	~370 emergent — redrill
4	NI 4-25	71.372995	38.738883	4313	600	588	12	320	~332 emergent — redrill
5	NI 5-25	71.375987	38.738293	4341	400	391	9	299	~308 emergent — **near melt-out**
6	NI 6-25	71.376728	38.735352	4343*	400	400	0	206	~206 emergent — redrill
7	NI 7-25	71.381125	38.739402	4387	400	391	9	271	~280 emergent — **near melt-out**
8	NS8-24	71.380200	38.730950	4450	200	—	—	—	Accumulation stake, reinforced; at the 2024 firn core site

* 2025 elevation differs ~40 m from 2024 at the same position — re-survey and flag. Naming: prefix was NS to 2024, NI from 2025; use NI n-26 and record whatever label is physically on the stake.

ASSUME EVERY STAKE NEEDS REDRILLING

Four of seven are projected within ~120 cm of melting out. In 2025 stake 1 was found **fully melted out** and had to be interpolated from stakes 2–6 — a lost observation at the most sensitive elevation. Carry enough 2 m segments to rebuild all seven, and **do the lowest stakes first** in case time runs short.

PROCEDURE

1. **Locate, read the physical label and record it verbatim** even if it disagrees with the expected label. Photograph as found with scale.
2. **Measure emergent length to the ice surface.** If the stake sits in a melt cup or on a pedestal, measure to the **general surrounding surface** and note the correction.
3. **Record tilt.** If it leans, measure the lean or record vertical height separately — an uncorrected lean is a systematic bias.
4. **Record a new GPS waypoint** — stakes move with the ice, so the 2026 position is itself a measurement.
5. **Drill the precise full stake length** (Kovacs 6 m auger, 1 m sections) so emergence at installation is zero — or record the exact insertion depth if not.
6. **If the old stake still holds, install the new one alongside and measure both.** Paired measurements are the only way to catch a bad reading.
7. **Photograph as left**; record new label, length and in-ice depth before moving on.

STAKE	LABEL FOUND	EMERGENT (CM)	TILT	NEW LABEL	LENGTH	IN ICE	NEW WAYPOINT	NOTES / PHOTOS
1								
2								
3								
4								
5								
6								
7								
8								

3 Glacier ice station (CR350)

Installed Sep 2025 by Stefan Fugger / Ines Dussaillant · Campbell CR350, USB-C · **2026 is the first data recovery** · condition unknown, possibly destroyed.

CONNECTING

Kit	The **same Toughbook as the AWS**. USB-C at the logger — **check the laptop's ports**, rugged laptops are often USB-A only, so you may need an **A-to-C** cable. Carry two.
Power	USB supplies 5 V and powers the logger for comms — **you can download even if the station battery is flat**.
Device type	**Select "CR300 Series" in the EZSetup wizard — there is no "CR350" in the list.** Expect the port as `CR350 series (COMn)`.
Settings	All defaults: PakBus address `1`, default baud, delays `0`, security code `0`, encryption key blank. **If a code was set in 2025 the logger will not respond and will look dead** — get it from Evan first.

NO SD CARD — AND THE MEMORY MAY HAVE WRAPPED

The CR350 has **no microSD slot**; the eject/green-LED routine from the AWS does not apply. Data is in internal flash, ~**50 MB**, and Campbell loggers default to **ring memory** — once full, the oldest records are overwritten.
Check the timestamp of the FIRST record, not just the last. That tells you whether the year is complete or the table wrapped. If the program used `FillStop` instead, logging stopped when full — which looks like a dead station but is a complete table.

DECIDE THE CASE ON ARRIVAL — DOWNLOAD BEFORE REPAIR, ALWAYS

CASE	WHAT YOU FIND	ACTION
A	**Standing and stable**	Download. Service, confirm logging resumes, leave in place. Record the depth register.
B	**Standing but loose or leaning** — mast melted out by a season of ablation	Download first. Then **redrill the mast** to depth and re-erect vertical. Record the new emergent height and full depth register — sensors are now at different depths than in 2025.
C	**Collapsed or destroyed**	Attempt data recovery regardless of physical state. **Photograph the failure mode thoroughly**, then **bring the station down off the glacier** — leave nothing on the ice. Budget the load capacity for this.

PROCEDURE

1. **Photograph as found** — wide and close. Record how much mast is emergent and whether it is vertical.
2. **Download via USB-C.** Verify continuity from installation to today; copy to two media before leaving.
3. **Measure emergent mast height and surface lowering** against the installation record. A year is ~2–3 m of ablation here. **This is what converts fixed sensor spacing into actual depths below the current surface** — it cannot be reconstructed later.
4. **Check every channel** returns plausible values; note any that failed, flatlined, or now read air because the surface dropped past them.
5. **Record battery state**; replace if there is any doubt about another year.
6. **Re-survey the position** — the station moves with the ice. Record displacement from 2025.
7. **Follow case A / B / C. Photograph as left** and record the full new depth register.

Date / time · **Case A/B/C**		Download successful?	
FIRST record timestamp		Last record timestamp	
Emergent mast at install / now		Surface lowering (cm)	
Battery (V) / replaced?		New waypoint / displacement	
Failed or suspect channels			
Sensor depth register (depth below current surface, per channel)			

4 Geoprecision bedrock logger (WP100) OPTIONAL

71.371561 E, 38.733798 N · 4416 m · installed Sep 2022 · probe in a 6 mm hole ~10 cm deep in bedrock. Record runs continuously since Sep 2022; peaked at 30 °C in summer 2025.

Model	GeoPrecision **M-Log5W** family, almost certainly **M-Log5W-Rock** — confirm before configuring.
Radio	**433 MHz** — why it reads at ~100 m, and why it penetrates snow well.
Hardware	**433 MHz USB dongle** — a hard dependency, no substitute.
Software	**GP5W-Shell** (older units FG2-Shell), Windows, free from GeoPrecision.
IDs	Logger `A542D3` · access code `DE10` · memory holds 30+ years hourly.

SET THE LAPTOP CLOCK TO UTC BEFORE READING OUT — AND DO NOT APPROACH THE SLOPE

Timestamps derive from the PC clock. Set it to **UTC, accurate to the second**, and **disable automatic daylight-saving adjustment**. Note this is the *opposite* convention to the AWS and CR350 — record which was used.
Read from the glacier at ~100 m. The block slope is moving and must not be entered. If the logger will not transmit, log that and stop — do not attempt physical access under any circumstances.

PROCEDURE

1. **Before leaving camp:** set and verify the laptop clock (UTC, DST off). Plug in the 433 MHz dongle and allow ~1 minute before launching GP5W-Shell.
2. **Approach to radio range from the glacier only** — ~100 m worked in 2025.
3. **Connect with access code** `DE10`; confirm the logger ID matches `A542D3`.
4. **"Load to Disk" → choose FULL, not Incremental.** Incremental fetches only new data. A full re-read costs minutes and makes this download self-contained.
5. **Save both formats** — compressed `.G2D`/`.F2B` and the expanded CSV.
6. **Verify before disconnecting:** record ends today and begins where the 2025 download ended; summer peak values plausible (30 °C in 2025 — a flat or clipped trace is worth catching now).
7. **Note battery level.** If a battery is ever changed: **save data to the PC first**, then re-verify every logger setting afterwards — settings can be lost with the power.

Date / time · read distance (m)		Logger ID confirmed	
PC clock UTC verified? DST off?		Full or Incremental upload	
Record start / end		Both formats saved?	
Battery level		Summer peak plausible?	
Notes / anomalies			

END OF DAY, EVERY DAY

Two copies of everything, on two media, carried by two people · GPX exported · field sheets transcribed tonight · photos renamed to the instrument they document · running list of anything broken, missing or deferred — that list is next year's equipment order.